

Numerical Solution of the

Robertson Chemical Kinetics Problem

Group E

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Presentation outline

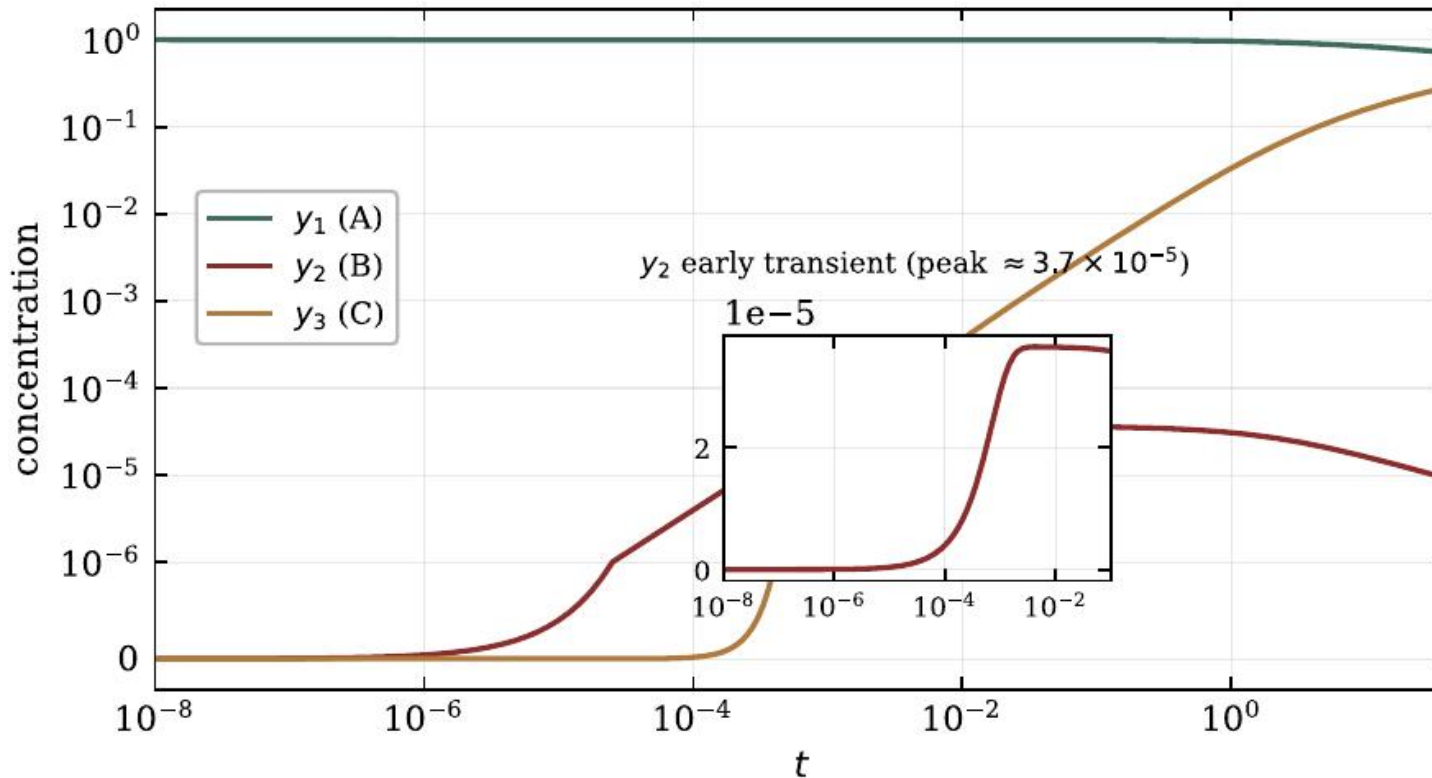
Background, model, methods, experiments and conclusions

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KEY TAKEAWAY **Course theory explains the solver behaviour observed in Robertson.**

Solving the Robertson problem

A stiff initial-value problem on $0 \leq t \leq 40$



Why do explicit solvers need tiny steps after the transient?

Three implemented methods

Explicit Euler (EE)
Classical RK4
Implicit Euler (IE)

Compare stability, endpoint error and computational cost.

KEY TAKEAWAY

A useful step must satisfy both stability and accuracy.

Three species, one conserved total

Course link: initial-value problems and invariants

Robertson reaction model

$$y_1' = -0.04 y_1 + 10^4 y_2 y_3,$$

$$y_2' = 0.04 y_1 - 10^4 y_2 y_3 - 3 \times 10^7 y_2^2,$$

$$y_3' = 3 \times 10^7 y_2^2.$$

$$\mathbf{y}(0) = (1, 0, 0)^T, \quad 0 \leq t \leq 40$$

Conserved total

$$y_1' + y_2' + y_3' = 0$$

$$y_1 + y_2 + y_3 = 1$$

Physical admissibility

$$y_i(t) \geq 0$$

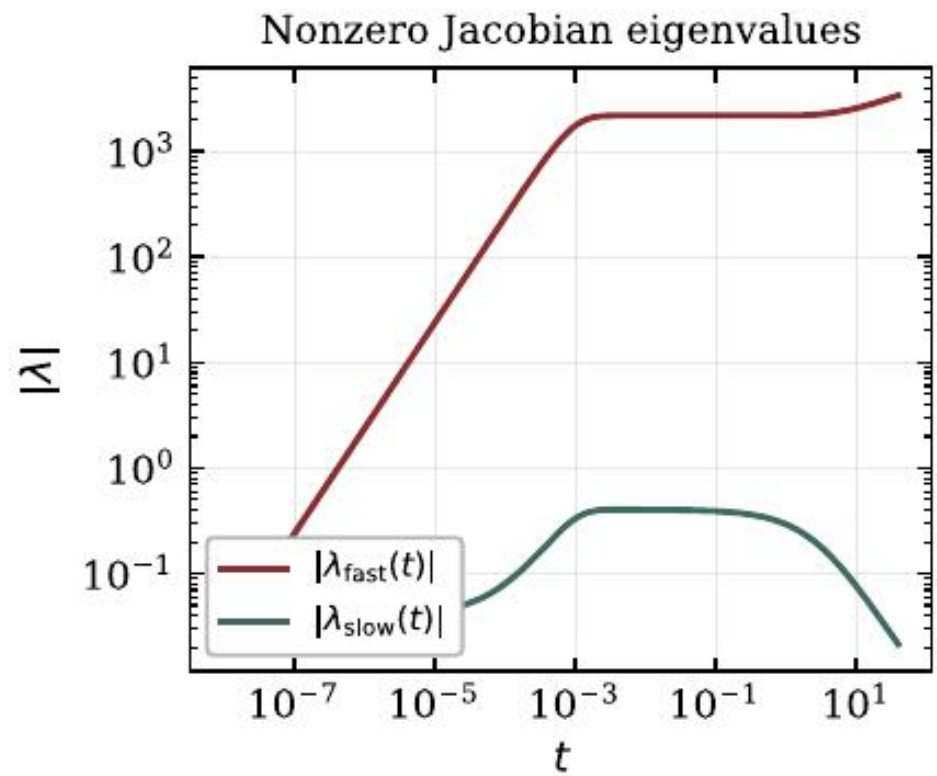
A conserved total alone does not prove accuracy.

KEY TAKEAWAY

Conservation and non-negativity provide separate physical checks.

The Jacobian reveals stiffness

Course link: linearization, eigenvalues and time scales



$$\delta \mathbf{y}' \approx J(\mathbf{y}) \delta \mathbf{y}$$

Local modes at $t = 40$

Local mode	Eigenvalue
Fast	-3392.79
Slow	-0.021419

$$S = \frac{|\lambda_{\text{fast}}|}{|\lambda_{\text{slow}}|} \approx 1.584 \times 10^5$$

Exclude the structural zero eigenvalue from conservation.

KEY TAKEAWAY **Fast modes restrict explicit steps even after the transient.**

Accuracy and stability are different properties

Course link: consistency, local error and global order

Method	Global order p	Work per step	Main strength
Explicit Euler	1	1 evaluation of f	Simple and cheap
Classical RK4	4	4 evaluations of f	Higher accuracy
Implicit Euler	1	Newton iterations	Stiff-mode damping

Local and global error

Local $O(h^{p+1})$

Global $O(h^p)$

The order does not determine the stability region.

KEY TAKEAWAY **RK4 raises accuracy order. Implicit Euler damps stiff modes.**

Newton and adaptivity solve different problems

Course link: implicit equations, Newton iteration and step doubling

1 Solve the implicit equation

$$G(w) = w - y_n - hf(w) = 0,$$

$$(I - hJ) \Delta w = -G(w).$$

Newton tolerance: 10^{-10}

Maximum: 12 iterations

If Newton fails, restore the saved state,
reduce h and retry.

2 Control the local time error

One full step $\rightarrow y^c$ (coarse),

Two half steps $\rightarrow y^f$ (fine).

$$\text{est} = \max_i \frac{|y_i^f - y_i^c|}{\text{atol} + \text{rtol} |y_i^f|}$$

Accept y^f if $\text{est} \leq 1$; otherwise retry.

$$h_{\text{new}} = 0.9 h \text{ est}^{-1/2}$$

Step-change factor clipped to $[0.3, 2]$.

KEY TAKEAWAY

Newton convergence and time-error control need separate checks.

Bounded trajectories can still be inaccurate

At $T = 40$, all listed runs are bounded and non-negative

Method	Step h	Steps	Endpoint error	Meets target?
EE	5×10^{-4}	80,000	5.938	No
RK4	5×10^{-4}	80,000	4.41×10^{-6}	Yes
IE	0.25	160	2938.9	No

Same step size

RK4 meets the accuracy target.

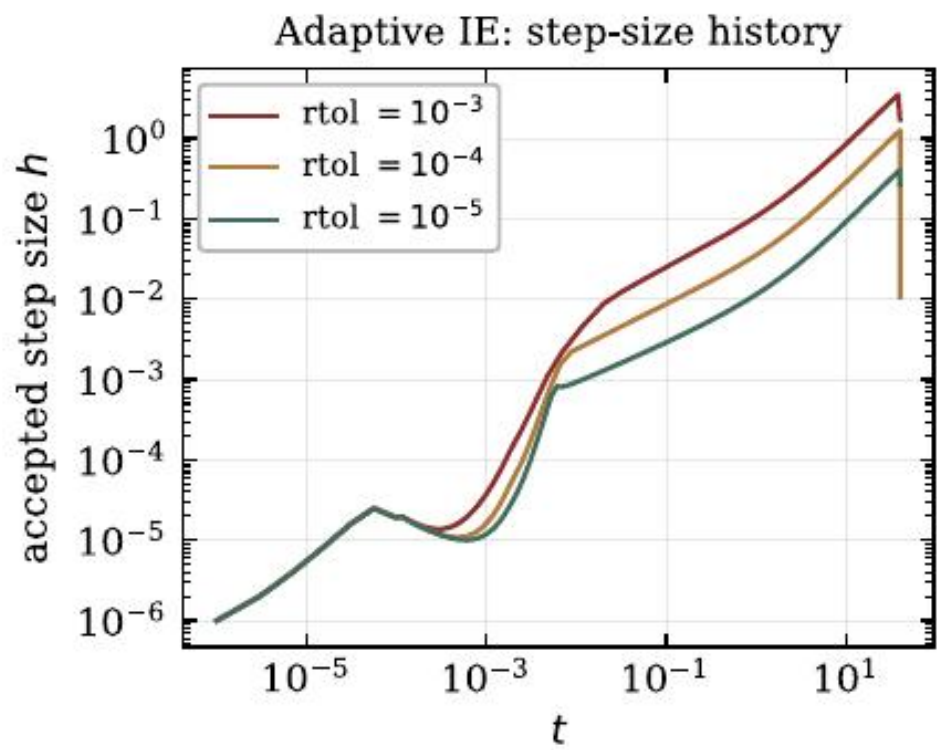
Larger implicit step

IE stays bounded but misses the target.

KEY TAKEAWAY Physical checks alone cannot establish numerical accuracy.

Local control is not an endpoint guarantee

Course link: local $O(h^2)$ control and first-order accumulated error



Adaptive IE with $\text{atol} = 10^{-8}$

rtol	Accepted steps	Endpoint error
10^{-3}	136	5950
10^{-4}	315	2010
10^{-5}	746	642
10^{-6}	1934	205

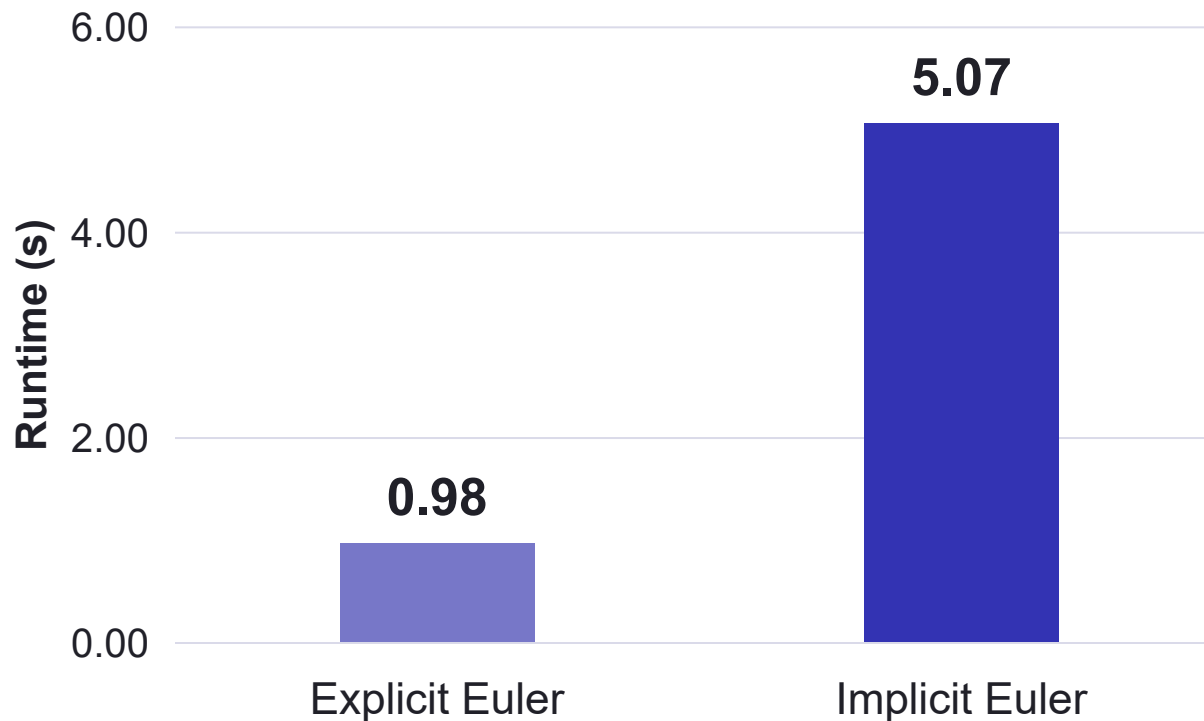
10× tighter rtol

About 3× smaller endpoint error in this sweep

KEY TAKEAWAY **Local tolerance control does not set the endpoint error.**

Efficiency depends on the accuracy target

Matched endpoint error: $E_T \approx 0.7423$



Same number of steps

640,000

Both first-order methods require small steps to meet this accuracy target.

IE adds a Newton solve to every step.

Separate work–precision experiment

External Radau: order 5 and L-stable.
Better trade-off in the tested range.

KEY TAKEAWAY

Compare runtime at matched error and include the work per step.

Solver selection: stability, accuracy and cost

Course concepts guide a practical solver choice

- 1 Stability** The Jacobian explains the small explicit step limits.
- 2 Accuracy** Implicit Euler damps stiff modes but remains first-order.
- 3 Cost** Compare runtime at matched error, including Newton work.

Next step Higher-order stiff methods and better adaptive control

KEY TAKEAWAY Combine stiffness analysis with error control and validation.